Digital Campus Library Reservation Gateway

Software Engineering Lab 1 | Requirements Engineering & UML Use-Case Modelling | Problem Statement #08

Submission Overview

This report consolidates the complete Lab 1 submission for the Campus & Academic Operations scenario. It includes the required seven-row requirements specification, the supplied hand-drawn UML use-case diagram preserved as an image, and the one-page UC-02 use-case flow.

1. Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a student member to search the ISBN catalog by title, author, or ISBN and view paginated availability results.	High	For a seeded catalog of 50,000 titles, each valid query returns matching records with title, ISBN, availability, and page controls; invalid/empty queries show a validation message.	Search is the primary entry point for locating library material.
FR-002	Functional	The system shall allow a student member to place a hold on a currently loaned book and assign the request the next FIFO queue position.	High	Given an unavailable title, the first valid request receives position 1 and later valid requests receive increasing positions; a member at the configured active-hold limit is rejected.	FIFO reservations provide fair access to scarce copies and enforce lending policy.
FR-003	Functional	The system shall notify the next eligible student member when a reserved copy becomes available and record the notification expiry time.	High	When a copy is returned, exactly the next eligible member receives an in-app and email notification within 60 seconds, with a configurable pickup deadline stored.	Timely notification converts queue movement into an actionable pickup.
FR-004	Functional	The system shall allow the head librarian to check out, return, and mark a reserved copy as collected or expired.	High	For each transaction, the system updates copy status, borrower/reservation state, and timestamp; a copy cannot be checked out twice concurrently.	Librarian operations maintain accurate inventory and reservation state.
FR-005	Functional	The system shall calculate, display, and record overdue fines for items returned after their due date.	Medium	For a returned item, fine = configured daily rate x overdue days; the recorded amount, member, item, and payment status are visible in the transaction history.	Fine records support transparent, auditable overdue handling.
NFR-001	Nonfunctional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms at the 95th percentile under simulated peak load.	High	A benchmark with 100 concurrent authenticated users records p95 response time below 150 ms for at least 1,000 representative queries, with no data-integrity errors.	Fast search preserves usability during peak academic demand.
NFR-002	Nonfunctional	The system shall protect member and transaction data using authenticated role-based access, TLS in transit, and audit logging for privileged actions.	High	Unauthenticated requests are rejected; students cannot invoke librarian actions; all librarian mutations create timestamped audit entries; TLS 1.2+ is enforced in deployment tests.	Library records contain personal and financial information requiring confidentiality and accountability.

2. UML Use-Case Diagram

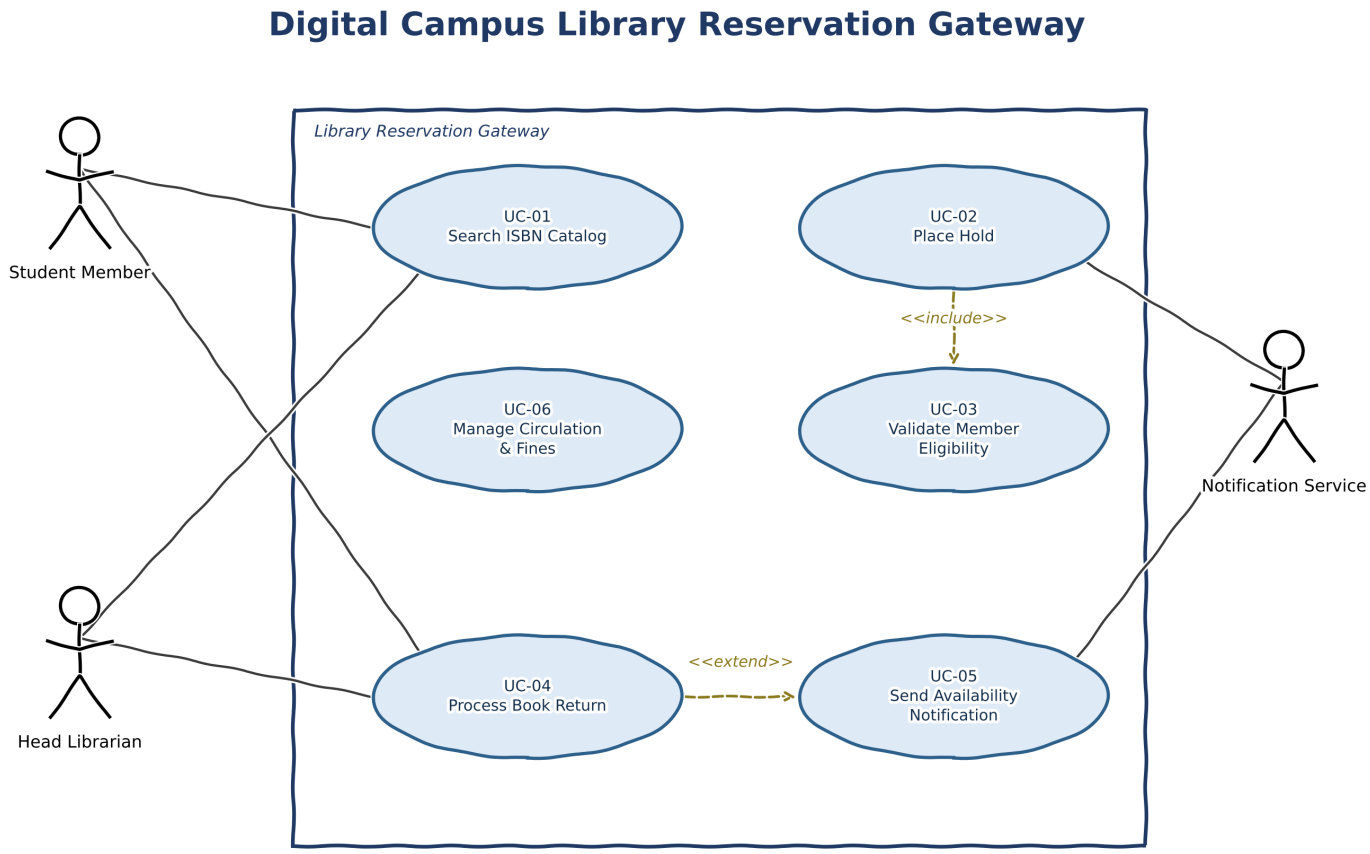


Figure 1. Supplied hand-drawn UML use-case diagram for the Digital Campus Library Reservation Gateway. The diagram shows the Student Member, Head Librarian, and Notification Service actors; six use cases; a system boundary; and the required include/extend relationships.

3. Use-Case Flow Specification - UC-02 Place Hold on Loaned Book

Primary actor: Student Member **Supporting actor:** Head Librarian / Notification Service **Goal:** Join the FIFO queue for a currently loaned book.

Preconditions

• Student is authenticated with an active library account.

• The ISBN exists in the catalog and all copies are currently loaned.

• The student is below the active-hold limit and has no duplicate active hold.

Postconditions

• A WAITING hold record stores member, ISBN, FIFO position, and timestamp.

• The member sees the assigned position and the librarian queue is updated.

• No existing checkout is changed.

Main Success Scenario

1. Student searches by title, author, or ISBN.

2. System validates the query and shows the loaned title.

3. Student selects the title and chooses Place hold.

4. System validates authentication, account standing, hold limit, and duplicates.

5. System locks the queue, assigns the next FIFO position, and creates the hold.

6. System releases the lock and schedules an availability notification.

7. System displays confirmation with queue position and notification conditions.

8. Use case ends successfully.

Alternate Flow - Member Not Eligible (4a)

4a1. If the limit, duplicate, or blocked-account rule fails, no hold is created.
4a2. The system shows the specific reason and required corrective action.
4a3. The student may return to search; the queue is unchanged.

Relationships

Place Hold **<<include>>** Validate Member Eligibility.
Send Availability Notification **<<extend>>** Process Book Return when a waiting queue exists.

Source deliverables: requirements_table.csv/docx/pdf, library_usecase_diagram_handdrawn.png, use_case_flow.md/docx/pdf, and the corresponding README traceability notes.